

Where does ReTrace find room for a watermark?

illustrative probability row

1 The model has local choice

several tokens remain plausible

The result MASK .

live conditional p_i



all are model proposals

2 Context assigns polarity

RRB measures a replayable response

two replayable context views

C^u: near context hidden

C^v: far context hidden

candidate	$g_i(z) = \log p^v - \log p^u$	
clear	-0.71	-
robust	+0.54	+
stable	+0.26	+
concise	-0.19	-
$q_- = .45$	$q_+ = .51$	$S_i = .90$



3 The key selects one side

RAR embeds; RRV replays

secret orientation asks for + response

RAR accepts from the requested set



emit "robust" • still a model proposal
RRV replay: $g_i(\text{robust}) > 0 \Rightarrow m_i = 1$

BALANCED



two-sided → carrier

ONE-SIDED



sharp → BCG skips

capacity = $\min(q_+, q_-)$